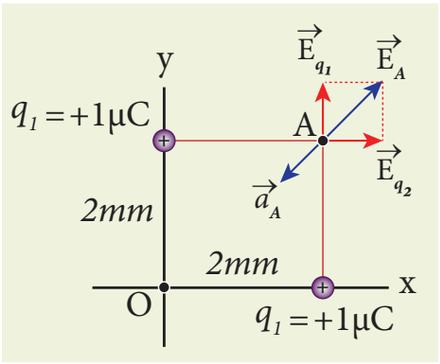
$= 2.25\times10^{9}\ \hat{j} + 2.25\times10^{9}\hat{i} = 2.25\times10^{9}\left(\hat{i}+\hat{j}\right)$

The magnitude of electric field

$$\left|\vec{E}_A\right| = \sqrt{\left(2.25\times10^{9}\right)^2 + \left(2.25\times10^{9}\right)^2}$$
$$= 2.25\times\sqrt{2}\times10^{9}\ NC^{-1}$$

The direction of $\vec{E}_A$ is given by $\frac{\vec{E}_A}{\left|\vec{E}_A\right|} = \frac{2.25\times10^{9}\left(\hat{i}+\hat{j}\right)}{2.25\times\sqrt{2}\times10^{9}} = \frac{\left(\hat{i}+\hat{j}\right)}{\sqrt{2}}$, which is the unit vector along OA as shown in the figure.



The acceleration experienced by an electron placed at point A is

$$\vec{a}_A = \frac{\vec{F}}{m} = \frac{q\vec{E}_A}{m}$$
$$= \frac{\left(-1.6\times10^{-19}\right)\times\left(2.25\times10^{9}\right)\left(\hat{i}+\hat{j}\right)}{9.1\times10^{-31}}$$
$$= -3.95\times10^{20}\left(\hat{i}+\hat{j}\right)N$$

The electron is accelerated in a direction exactly opposite to $\vec{E}_A$.

1.3.3 Electric field due to continuous charge distribution

The electric charge is quantized microscopically. The expressions (1.2), (1.3), (1.4) are applicable to only point charges. While dealing with the electric field due to a charged sphere or a charged wire etc., it is very difficult to look at individual charges in these charged bodies. Therefore, it is assumed that charge is distributed continuously on the charged bodies and the discrete nature of charges is not considered here. The electric field due to such continuous charge distributions is found by invoking the method of calculus.

Consider the following charged object of irregular shape as shown in Figure 1.9. The entire charged object is divided into a large number of charge elements $\Delta q_1, \Delta q_2, \Delta q_3 \ldots\ldots \Delta q_n$ and each charge element Δq is taken as a point charge.

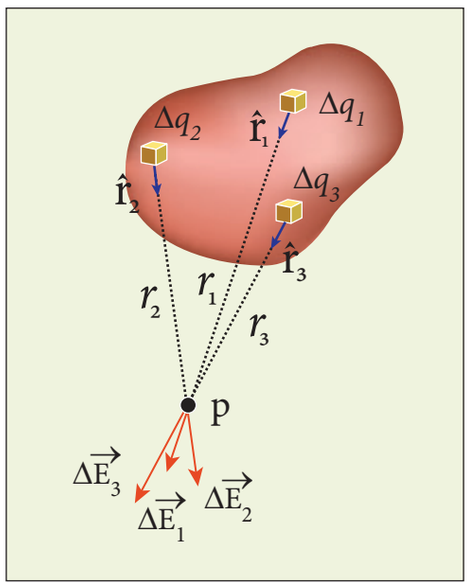


Figure 1.9 Continuous charge distributions

The electric field at a point P due to a charged object is approximately given by the sum of the fields at P due to all such charge elements.

$$\vec{E} \approx \frac{1}{4\pi\varepsilon_0}\left(\frac{\Delta q_1}{r_{1P}^2}\hat{r}_{1P} + \frac{\Delta q_2}{r_{2P}^2}\hat{r}_{2P} + \ldots\ldots\ldots + \frac{\Delta q_n}{r_{nP}^2}\hat{r}_{nP}\right)$$
$$\approx \frac{1}{4\pi\varepsilon_0}\sum_{i=1}^{n}\frac{\Delta q_i}{r_{iP}^2}\hat{r}_{iP} \qquad (1.9)$$